

ADDIS ABABA UNIVERSITY

ADDIS ABABA INSTITUTE OF TECHNOLOGY - AAiT

SCHOOL OF INFORMATION TECHNOLOGY AND ENGINEERING – SiTE

Guideline for MSc Thesis Proposal Writing and Evaluation

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1. Preface

This document is prepared to provide guidelines for MSc students on how to write MSc research proposal. It also gives the necessary information on how a research proposal is submitted, evaluated, and approved at School of Information Technology and Engineering (SiTE).

It is the student's responsibility to read and fulfill the requirements presented here and to submit a document of high quality dissertation proposal. Those proposals that do not fully comply with this document shall not be accepted.

2. Introduction

A research proposal is a short document designed to express the researcher's research problem, method and plan to conduct a research. It can also be defined as a reasoned, critical research plan that includes problem definition and how researchers will handle them.

An approved research proposal is an intellectual academic contract between the student and the School Graduating Committee (SGC). It specifies what is exactly expected from the student and this specifically allows the SGC to evaluate the student based on what is originally proposed.

Once a student chooses a topic for research in any area related to the MSc program and a supervisor is assigned, the details of the research proposal should meet the following guidelines:

- A written document including: a research plan (research questions and methods), a thorough literature review, a timetable, and a statement of the resources required to complete the research (detail contents are available below).
- A brief oral presentation.

The proposal evaluation committee and Members of the SGC, having attended the presentation and examined the written proposal, will give comments, recommendations, and their respective evaluations accordingly.

3. Detailed outline for the Research Proposal

The MSc proposal should be divided into different sections as shown in table 1; the length of the page must not exceed 25. Interline spacing should be 1.5, and of 12 point type.

The structure and content of the graduate research proposal should include the following sections as shown in table 1.

Table 1: Graduate research proposal contents

Preliminary pages	• Title page • Approval page • Abstract • Acknowledgment (optional) • Table of content • List of figures • List of tables • List of acronyms
Main body of the proposal	1. Introduction 1.1. Background Information 1.2. Motivation of the Study 1.3. Statement of the Problem 1.4. Research Questions 1.5. Objective of the Study 1.6. Expected Contribution of the Study 1.7. Scope/delimitation 1.8. Structure of the Document 2. Literature review 2.1. Literature review 2.2. Related works 3. Methodology 4. Work Schedule and cost breakdown 5. References 6. Appendix 7. Assurance

3.1 Title Page: Give a title to the research proposal. The title should be concise and clear. Use few words to adequately describe the contents of the study. From the title, the reader should be able to predict fairly and accurately what the research will be about.

The title page is the first page of the proposal and it has no page number.

The title page should include the following:

- AAU, AAiT logo
- Full title of the proposal
- Full name of the student
- Full name of advisor(s)
- Name of the department
- Name of the school
- Name of the institute
- Name of the university
- Month and year of submission

3.2 Approval Page: the approval pages come after the title page. The details are available in the sample thesis proposal format.

3.3 Abstract: is a one page brief summary of the thesis proposal. It should inform the readers why a particular topic is important? How the research is going to answer the research questions? How the work fits into what is already known about the research topic? And what novel contribution the research will bring about? Even though it appears at the front of the proposal, it is written last.

3.4 Table of Contents: The tables of content list in sequence all relevant subdivisions of the proposal with their corresponding page numbers. It includes:

Approval page

List of table (if any)

List of figures (if any)

Abbreviations and acronyms (if any)

Abstract

1. Introduction
    1.1. Background Information
    1.2. Motivation of the Study
    1.3. Statement of the Problem
    1.4. Research Questions
    1.5. Objective of the Study
    1.6. Expected Contribution of the Study
    1.7. Scope/delimitation
    1.8. Structure of the Document
2. Literature review
    2.1. Literature review
    2.2. Related works
3. Methodology
4. Work Schedule and cost breakdown
5. References
6. Appendix
7. Assurance

3.5 Introduction/Background Information: In this section, the student should provide background information for the research proposal. The background should engage readers with broad themes and topics. At one level, the purpose of a background section is to give the reader the relevant facts about the research topic so that readers understand the material that the candidate is writing about and how it links to his/her research question. This section must not, however, simply provide the general context, but must direct the readers' attention to the empirical details through which the research topic and question are made relevant. As such, this writing must not just fill in details of the topic the candidate is researching, but implicitly illustrate the need for and importance of the research.

3.6 Motivation of the Study: In this section, the student should provide what are the motivations for conducting this MSc thesis work.

3.7 Statement of the Problem: A problem might be defined as the issue that exists in the literature, theory, or practice that needs further study. It is important in a proposal that the problem should be defined clearly so that the reader can easily understand the problem that the candidate is going to solve. Statement of the problem summarizes the question the researcher is trying to answer. Effective problem statements answer the question "Why does this research need to be conducted?" It establishes foundation for everything to follow in the proposal. Therefore, it should be written clearly and completely.

3.8 Research Questions: Questions and propositions summarizing the applicant's expected findings in the proposed research, and presenting a clear rationale for each hypothesis or research question. (In case of studies involving quantitative approach, for every hypothesis, variables to be measured/tested should be clearly stated).

3.9 Objectives: The objectives of a research define the ends or aim which the researcher seeks to bring about. An objective may be thought of as either a solution to a problem or step along the way towards achieving a solution. Objectives of a research summarize what is to be achieved by the study; it should be closely related to statement of the problem; It should be simple (not complex); specific (not vague); and it shall be stated in advance (not after the research is done). Objective should be written in two forms namely general objective and specific objective.

3.10 Literature Review: Brief description about the research problem should be presented in this section. It should mainly focus on the already existed science towards the research problem.

3.11 Related Work: the most relevant studies around the topic of research need to be discussed under this sub-section to show how the topic at hand is similar with other previous works, yet at the same time different because it deals with a unique and unaddressed topic of interest. It is in this section that the applicant **comprehensively reviews the literature** pertinent to the problem to show what other researchers have published on the problem (citing specific authors where appropriate), what **gaps** of knowledge still exist, and what **additional research** needs to be done. The research problem has to be delineated or identified

in this section. It is also in this section that the candidate demonstrates his/her mastery of the theoretical subject matter in the research area.

3.12 Significance/Contribution of the Study: In this section, indicate how the research will refine, revise, or extend existing knowledge in the area under investigation. Elaboration of the importance of the study and advantages to be derived showing how this study will add to the body of knowledge. The reader has to be convinced that the proposed research addresses an important clearly defined question pertaining to topic at hand. Here the contribution that will be expected from the MSc research study shall be stated.

3.13 Scope/Delimitation of the Study: the scope refers to what factors/aspects are attended to and are within the accepted range of the research study. All research has limitations and thus certain work that **will not** be performed should be written. The work that will not be undertaken is described as the ***delimitations*** of the research.

3.14 Proposed Methods: A dissertation proposal should contain a section giving details on the methods, approaches and techniques to be used when conducting the research. The location(s) where the proposed research will be carried out (when applicable) should also be given in this section. If particular instruments are to be used, their details and specifications should be presented. If the data are to be collected through sampling, then the research design and sampling procedures should be described. If questionnaires will be used, samples of the proposed questionnaires should be presented. If standard methods will be used, then full references to them should be given. In all cases, the data analysis plan should also be included. If any limitations to the proposed methods are known, then these should be pointed out.

Specifically this section is divided into:

- **Study design:** What approach will you take to answer your research question or hypotheses and why? The method must be consistent with the stated research question.
- **Sample:** What population have you chosen? and why? Define the inclusion criteria and exclusion criteria for your sample. State how you will recruit participants to be in the study and what sampling method you used. You also need to state how the sample size was determined.
- **Instruments:** what instruments are you going to use to gather data? Report the validity and reliability of each instrument.

- **Variables:** (for quantitative approach)**:** Define the variables of interest for the study and provide both conceptual and operational definitions. Explain why you believe the tools used to measure variables are valid and reliable. Specify the Independent, dependent, intervening and extraneous variables.
- **Proposed Hypotheses: Hypotheses** are a tentative, intelligent guesses to the solution of a research problem that can be validated through experiment, theory or other research methods. If there are any temporary solutions for all the sub problems that are mentioned in the previous sections, they should be presented in detail in this section.
- **Evaluation mechanism** (for design science) - discuss how the proposed design will be evaluated.
- **Procedure:** How will you actually conduct the study? Explain exactly what data will be collected, how they will be collected and who will collect them and what framework is used.
- **Data analysis:** there must be consistency between the data obtained and the type of analysis selected for the study. There also should be a consistency between the research questions and the analysis used to answer those questions.
- **Ethical concerns:** discuss what ethical issues need to be considered; for example, how you will protect participants' confidentiality.

3.15 Work Plan: gives a brief outline of the estimated time needed to complete each section of the research. In this section, if the researcher makes any progress in the research study such as publications, experimental output and other, he/she can present it.

3.16 Budget: this section should give the source of funds required for the proposed research and also the proposed budget.

3.17 Timetable: this section should give the total time within which the research is expected to be completed, and also a time-table showing the sequence of the research activities.

NB: Work Plan, Budget and Time table should be annexed

3.18 References: all the literature that has been referred to in the text must appear in the reference list. In writing references / bibliography researcher must follow the standard guidelines given by IEEE, Chicago, Harvard, etc. (for consistency purpose, It is recommended to use IEEE format). In which the details required to cite the reference are: the name of the author(s) starting with surname, year of publication, title of the book, edition, place of

publication, name of the publisher(s) (in case of books. (Please refer to specifics on APA style online, Google scholar).

While writing reference, follow the following rules:

- Any work or statements not your own should be clearly referenced.
- Every reference in the text should be available in the reference list.
- Every article in the reference list must have a reference in the text.
- Every table, figure, or photograph must have a reference in the text.
- Follow a specific and consistent guideline regarding use of references in text and in the reference list.

3.19 Assurance: The student and advisor(s) should agree to accept responsibility for the scientific, ethical and technical conduct of the research by affixing their names and signatures.

4. Procedure for Preparation and Approval of Graduate Research Proposals

The procedure for preparation and approval of the MSc proposals is given below.

1. Research advisors must be assigned and approved for each student by SGC. Students may also recommend advisors. However decision shall be made by SGC.
2. Two supervisors may be assigned to guide a candidate through the MSc research study. Where such two supervisors are assigned, one of the two shall be designated as the *Principal Supervisor* (*Advisor*) and the second one as *Co-supervisor* (*co-Advisor*), which optional. The requirement for MSc thesis advisor is i) having assistant professor rank and above in related filed of the study from AAU as a *principal Supervisor*. However, co-advisors with Assistant Professor rank and PhD holder in related field of the study shall be assigned from AAU, other universities (local or abroad), or from the industry.

1. In consultation with his/her advisor(s), the student identifies a research topic and develops a master's thesis proposal which addresses the following points that are presented up there: nature and significance of the problem, expected objectives and outcome, description of the procedures used to solve the problem, timetable for the work and budget breakdown and etc.
2. Advisor(s) critically review the MSc thesis proposal and check whether the student's proposal is up to the standard and prepared following this guideline.

3. The student shall submit three hard copies and soft copy, signed by his/her advisor(s), to the school.
4. Following the SGC meeting on the submitted proposal, SGC chairperson passes a copy of the MSc thesis proposal to examiners (both internal and external) who are expert in the area and arranges proposal defense presentation.
5. SGC should approve the examiners.
6. Assigned examiners/evaluators shall thoroughly review the MSc thesis proposal and evaluate using ***master's thesis proposal evaluation*** form for the MSc program. After which it may a) return the proposal to the student for clarification, modification, rewriting or rework, or b) approve the proposal as submitted. (the detail evaluation procedures are presented in ***Evaluation procedure for the thesis research proposals*** section.5)
7. Student makes corrections to the accepted proposal and submits 4 signed copies of the corrected proposal (with signatures of evaluators and advisor(s) on the approval page of the proposal) to the school after advisor(s) checked that all agreed corrections and suggestions of assigned evaluators are included. Students with re-work proposals should follow procedures 1-6 again.
8. After checking the inclusion of all corrections and suggestions, the SGC chairperson duly signs the approved copies of the proposal and sends it to SGC.
9. The SGC will assess the proposal in terms of the guidelines used and feasibility including the cost breakdown.
10. Following the SGC decision, the School Dean/PG coordinator will approve the proposal by duly signing each copy. One copy of the proposal should be sent to the office of Director for Graduate Program (DGP).

NB: As per the guidelines and regulations of AAU, AAiT, if a student does not have an approved MSc thesis proposal, then it is taken as the student is not making "Normal Progress". The SGC and AC must be made aware of the reasons for the delay, expected timeline for completion, and a meeting may be requested with the student and advisor to discuss the circumstances.

5. Evaluation Procedure for the Thesis Research Proposals

The student will first submit his/her proposal to the school, **three weeks before** the scheduled proposal presentation date, both in hard and soft copy. The School shall distribute the proposal to the research committee members (SGC).

Students will be asked to present their work to the committee members according to a schedule the school will announce. All members of the committee have an equal opportunity to express their concerns and provide their comments in the evaluation process. Members must exercise their independent judgment having in mind the best interest of students, as well as their background, abilities and qualifications. However, the student will be evaluated by two internal and one invited external examiners.

The master's thesis proposal shall be evaluated based on the "The Checklist for Research Proposal Evaluation" and students' standing shall be described as ***Pass***, ***Pass pending modifications (PPM)*****,** or ***Fail (F)***. In the event where a student earns a grade of ***F***, s/he would be asked either to revise the proposal or come up with new proposal and present it again.

The examiners are expected to communicate decision, along with comments, to the student **in writing within a week** following the evaluation meeting. The student should review the evaluation form with the supervisors. The student is expected to address the comments and points raised by the committee to obtain final approval and signature of the supervisor based on the committee's decision. During revision of proposals, any future major divergence from the originally presented proposal should be communicated to the research committee for review and approval before the student continues working on the thesis research. Normally, final approval should be obtained within one or two weeks of completing all required revisions.

During the proposal presentation, the student and the SGC are encouraged to involve other faculty members from outside the committee, or the program, for additional input and suggestions.

6. Procedures for Oral Presentation of the Proposal

The chair of the SGC will begin the meeting with introductions and steps of the meetings.

The student will present for 20 minutes.

The examiners committee will then ask the student questions regarding the research study for about 20-30 minutes each.

The student will be asked to leave the room so that the committee can discuss the oral presentation and written document.

The examiners and (optionally the supervisor) will then complete the checklist for research proposal evaluation while the student is outside of the room.

The supervisor will then ask the student to return to the room and the student will be informed of the result and what the next research steps will be at the point.

Copies of the completed Research Proposal Evaluation form should be submitted to the school dean office as well as the graduate program coordinator.